HON207
Cognitive Science

Sequential Sampling Models

Choice RT

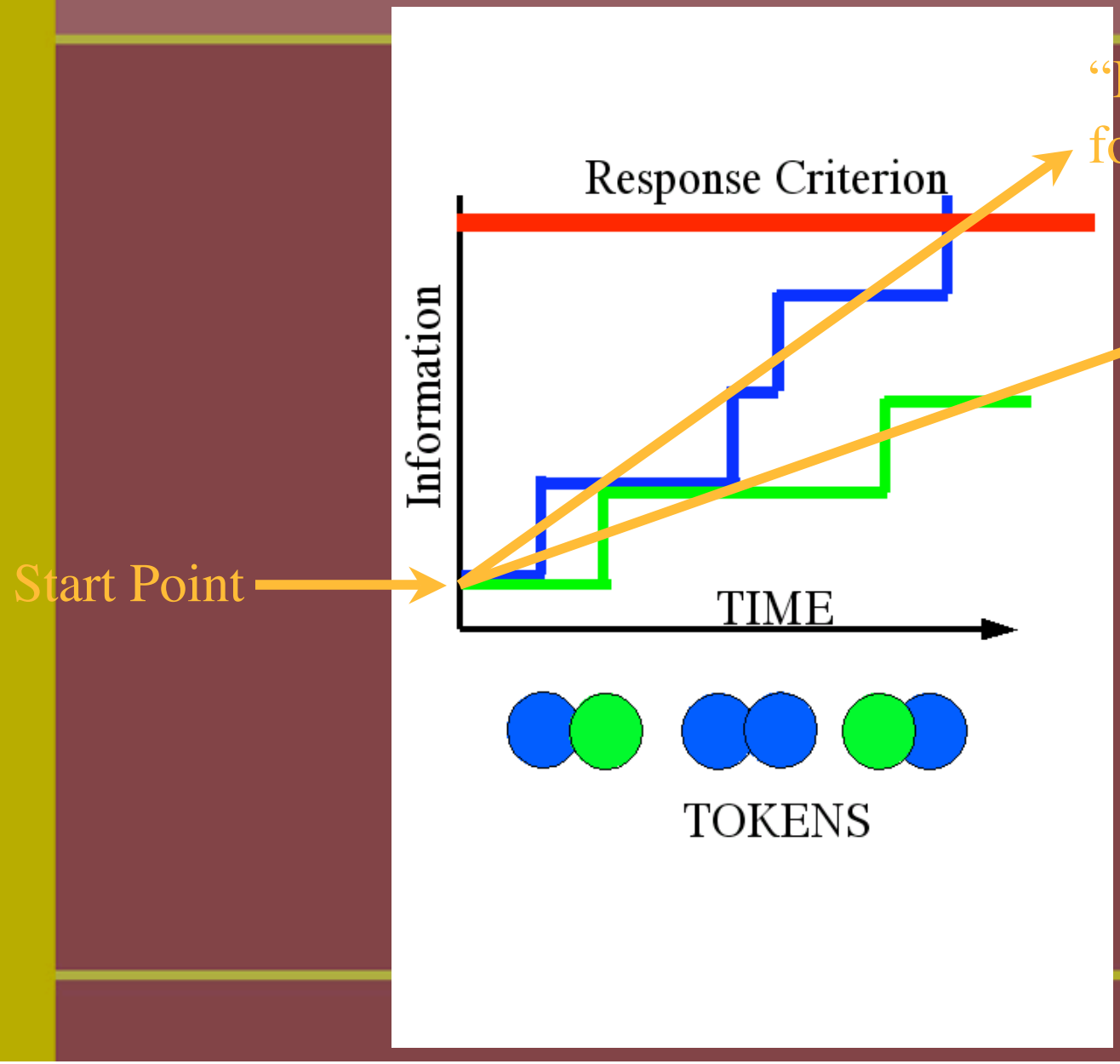
- Response time is a rare measure in psychology.
- Choice vs. simple RT.
- Amenable to computational modeling.
- Long history
- Decision making is a wide field, RT is a useful by-product of decision making.

2AFC

- Two-alternative forced choice.
- Models most memory research, some categorization research, lexical decision, etc.
- Well studied in the choice RT domain.
- More than two choices, though?

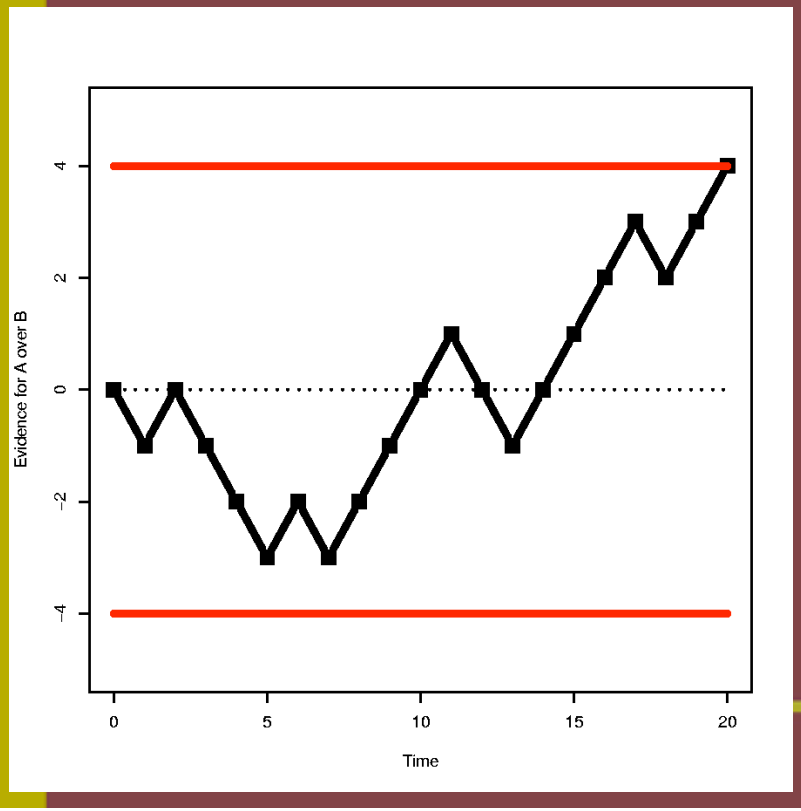
Sequential Sampling

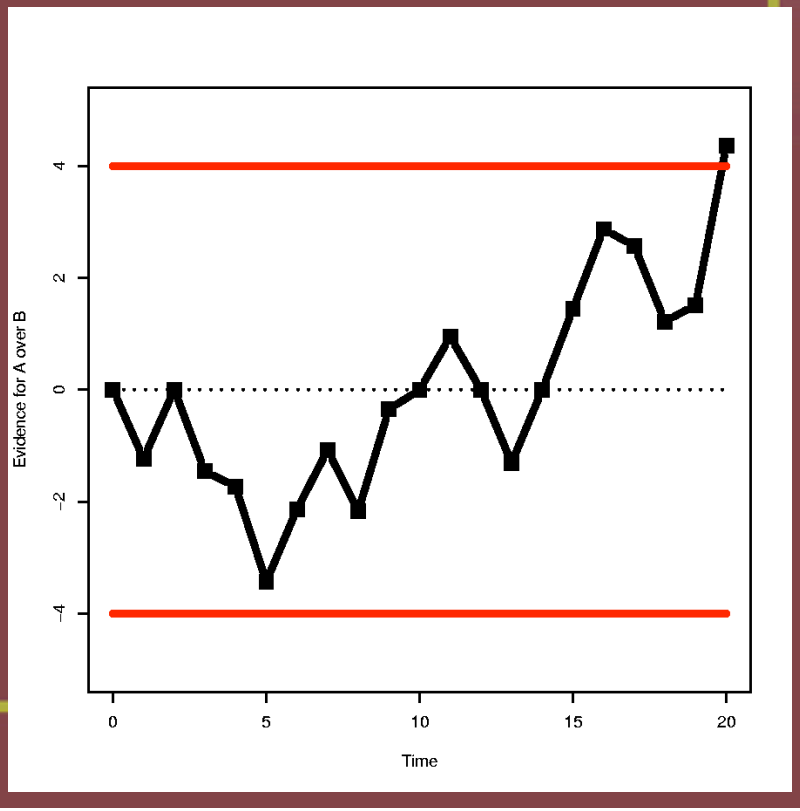
- You need to decide on response A vs B.
- Collect a discrete, small piece of evidence that favors A or B (sampling).
- Repeat until you have enough evidence (sequential).
- Naturally predicts RT (how many samples?) and accuracy (what was the final decision?).


"Drift rate"
for blue
Response Criterion
"Drift rate"
for green
Information
Start Point
TIME
TOKENS

Variations

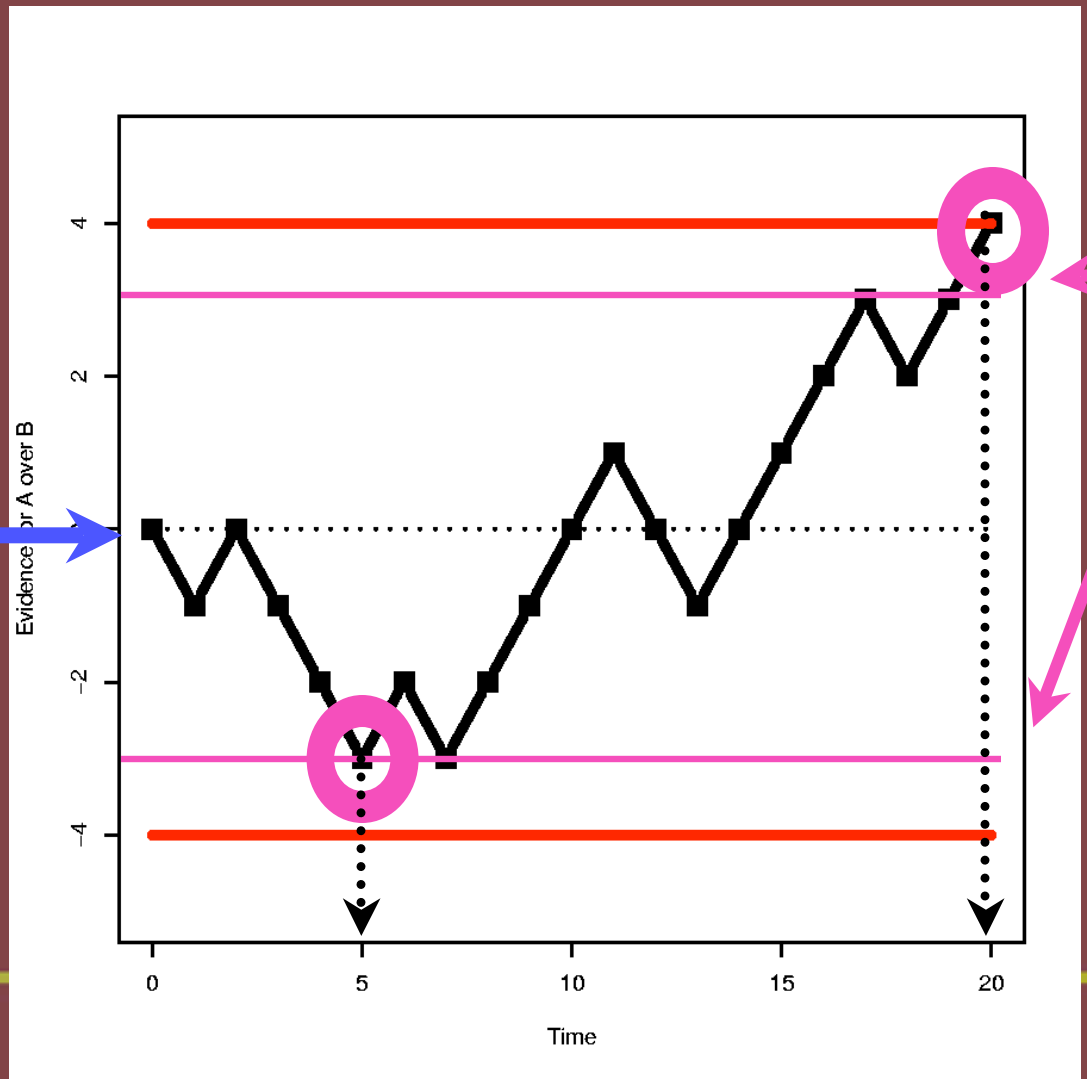
- That was discrete time and discrete state.
- Either/both could be continuous.
- E.g., random walk models:





Psychology

- Links between parameters and psychology.



Sources of Noise

- All good cognitive models include variability.
- SS models have variable evidence (discrete or continuous).
- Possibly also variable arrival times.
- This is not enough:
  - RT distributions are identical for correct and error.
- Empirically, there are:
  - Fast errors in easy conditions with speed emphasis.
  - Slow errors in hard conditions with accuracy emphasis.

More Variability

- Adding between-trial variability changes error RT distributions.
- Variability in bias (start point) -> fast errors:
  - “Jump out” errors, where you’re biased to begin with.
- Variability in drift rate -> slow errors:
  - Slow, unsure responses with high error rates.

Neural Accumulators

- Very promising.
- Single cell recordings agree.
- Neuro-versions have almost no parameters.
- Slowly bridging the gap from action potentials to behavior.

Analytics

- Valuable, for:
  - Fitting & parameter estimation.
  - Intuition.
  - Completeness.
- But difficult:
  - Requires stochastic d.e. theory.
  - Mostly Ito calculus.
  - Nasty, not easily extensible.

Computation

- Incredibly simple.
- Sequential sampling is a natural loop.
- Sample random numbers (tokens of evidence) until a criterion is reached!
- Repeat many times.
- Separate correct & error trials, draw histograms.

E.g. Simple Accumulator.

1. Two counters, A and B. Initialize at A=0=B. Time=0.
2. Sample $x \in [0,1]$ random & uniform.
3. If x<drift rate, A=A+1, else B=B+1.
4. If A or B > criterion, goto 6.
5. Time = time+1. Goto 2.
6. If A>crit, response is A, else response is B.

Computational Variations

- Continuous time:
  - Step 5: time = time + (random amount)
- Continuous evidence:
  - Step 3: A=A+(random) or B=B+(random)
- Response bias:
  - Step 1: A=5 (say) and B=0.
- Start point noise:
  - Step 1: A=random, B=random.
- Drift rate variability:
  - Each decision time, chose drift rate randomly.